S&P GLOBAL & CRISIL CAMPUS HACKATHON 2026 · AI/NLP FINANCIAL RISK ENGINE

Signal Risk

From unstructured news, filings and posts to structured risk signals, and from those signals to an index that rebalances on sentiment and a bank that stress-tests itself when the world changes.

Candidate
Naman Karwa

College
IIT Guwahati

Modules built
Engine · A · B

PROBLEM & APPROACH

Markets move on text long before it becomes data

A sanctions headline, a downgrade, a bank run: risk teams read them hours or days before the information reaches structured feeds and the next scheduled stress test.

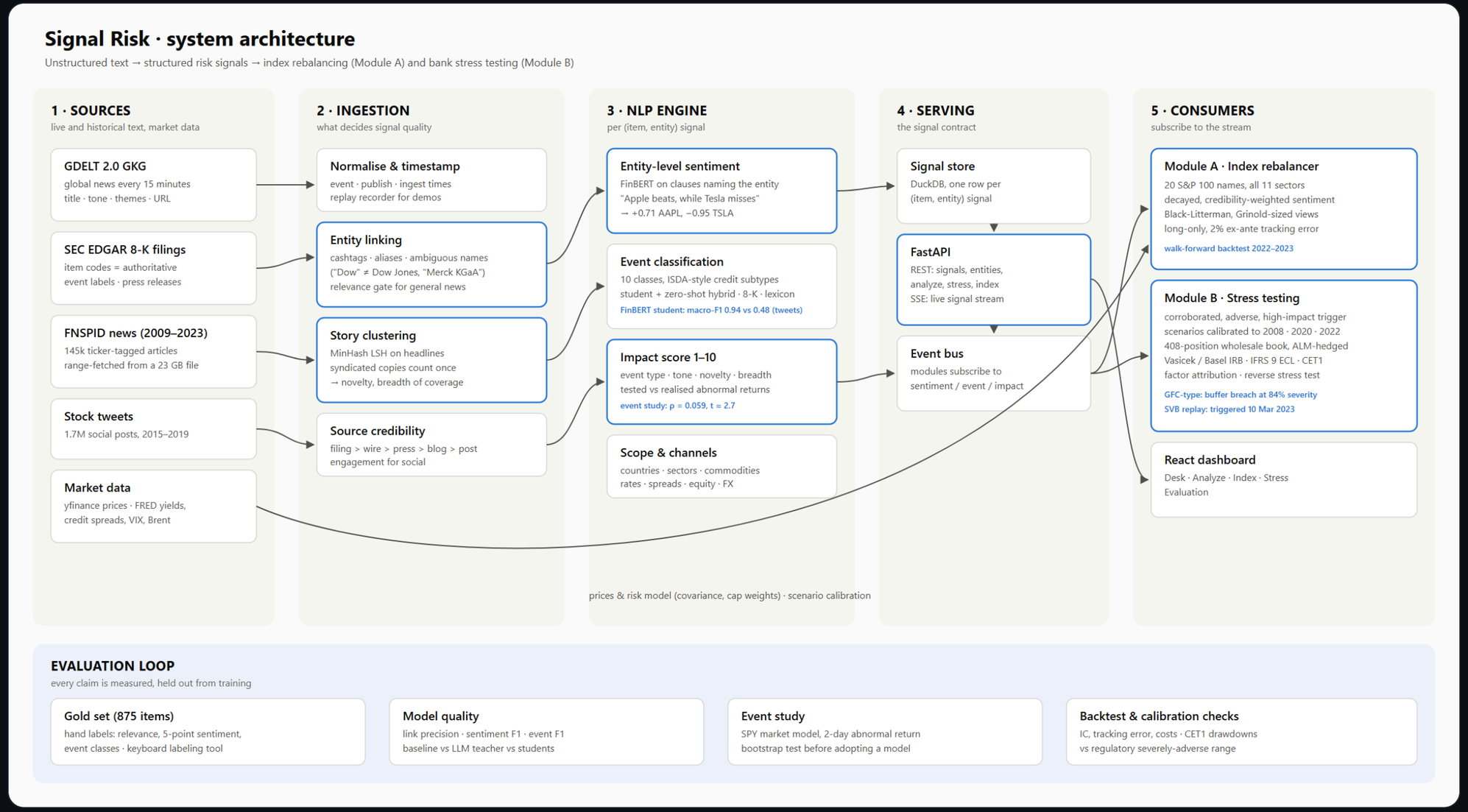
The hard part is not running a sentiment model. It is **knowing which company a sentence is about**, **counting a syndicated story once**, **telling chatter from events**, and proving that an "impact" score means something in market terms.

Approach

- **Engine:** per-entity sentiment, 10-class events with ISDA-style credit subtypes, impact 1–10, plus the scope and transmission channels downstream risk needs
- **Module A:** Black-Litterman index tilted by sentiment, under a 2% tracking-error budget
- **Module B:** wholesale bank book stress-tested on corroborated adverse events; CET1 under IFRS 9 and Basel IRB
- **Every claim measured:** event study, walk-forward backtest, gold set, calibration to 2008/2020/2022

SYSTEM DESIGN

Five stages, one signal contract, an evaluation loop around all of it



IMPLEMENTATION HIGHLIGHTS

Where the engineering went, and why

Entity linking that understands finance

Cashtags, aliases and guarded ambiguous names: "Dow" is not Dow Jones, "Merck KGaA" is not Merck, an apple pie is not Apple. Clause targeting gives "Apple beats, while Tesla misses" **+0.71** and **−0.93**.

Stories, not articles

MinHash clustering counts syndicated copies once and turns breadth of coverage and novelty into features; a relevance gate keeps general news honest.

Student over zero-shot

FinBERT fine-tuned on 12.8k finance tweets: macro-F1 **0.94 vs 0.48** for zero-shot on de-duplicated held-out tweets; on news the gain is smaller (0.54 vs 0.49). Hybrid keeps zero-shot for classes without data; a Groq-hosted LLM teacher fills them.

Data where it was hard

GDELT's 15-minute GKG files (the search API throttles), SEC 8-K item codes as free event labels, 145k FNSPID articles range-fetched from a 23 GB file (6% downloaded).

Portfolio construction

Sentiment views sized by Grinold's rule ($\alpha = IC \times \sigma \times score$) inside Black-Litterman, Ledoit-Wolf risk, long-only, 2% ex-ante TE, no-trade band.

Bank-grade stress engine

Vasicek conditional PDs, IFRS 9 stage 2 lifetime ECL, Basel IRB RWA, revolver drawdowns, ALM-hedged rate book, factor attribution and reverse stress testing.

Python · PyTorch (CUDA, laptop RTX 3050) · Hugging Face Transformers · DuckDB · FastAPI + server-sent events · React · ECharts · 65 automated tests

KEY RESULTS

Measured, not asserted

84%

of Lehman-2008 severity breaks the bank's 8% CET1 requirement (90% for March 2020; a Russia-Ukraine shock never does, even at 150%)

10 Mar 2023

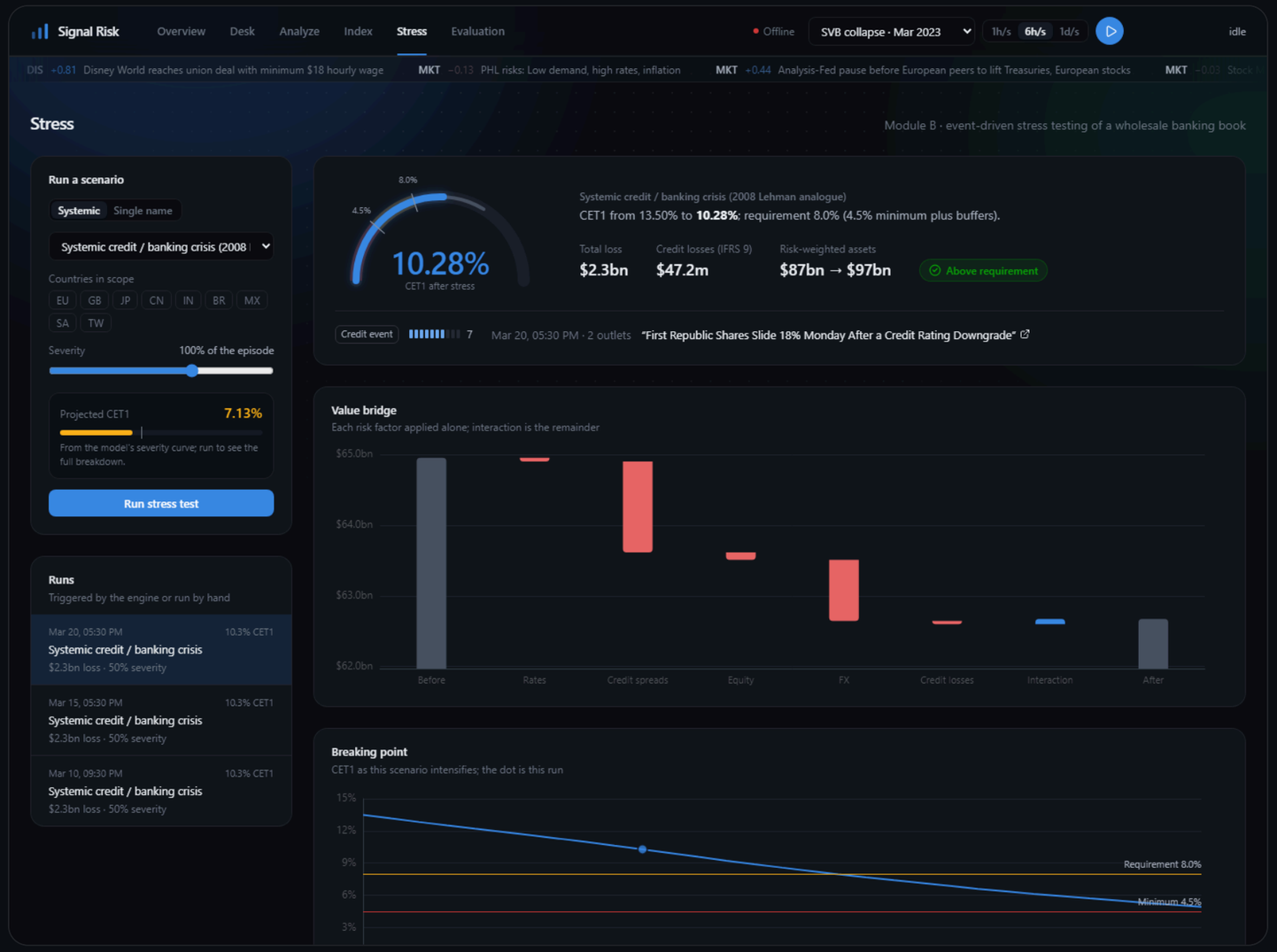
SVB replay: systemic stress test triggered on "Silicon Valley Bank's Liquidity Crisis"; re-run for Credit Suisse and First Republic; zero false alarms on quiet news

ρ = 0.059

impact score vs realised 2-day abnormal returns on 2,141 held-out stock-days (≈ 2.7 SE); a boosted model was not adopted because its gain was not significant

2.1% of 2% budget

Module A realised tracking error, 2022–23 walk-forward with costs; +1.1% p.a. active (t = 0.74)



DOMAIN IMPACT

From a headline to a capital number in minutes

For a bank's risk team: a corroborated adverse event becomes a scenario calibrated to history, scaled by its severity and targeted at the countries it names, and lands as CET1 headroom, the exposures driving the loss and the severity at which buffers break. Stress testing stops being a quarterly ritual.

For a portfolio manager: sentiment enters as a sized, confidence-weighted view under a hard risk budget, so a burst of credible news tilts the index and a single viral post does not.

For model risk: every number has a test behind it, and the system says when it is not good enough.

Findings a desk can use

- **News is priced within a day:** sentiment agrees with the same-day move ($\rho = +0.086$) but reverses slightly when traded a day later (IC -0.024). Speed matters, so the live index rebalances on 15-minute news.
- **Media attention is not market impact:** 193 outlets covering a political sanctions story triggered nothing once breadth was damped and economic content required.
- **Capital is most exposed to credit and pandemic-type shocks,** not geopolitical ones, for this book.

LIMITATIONS & NEXT STEPS

What is honest to say, and what comes next

Limitations

- Impact is a heuristic score; it ranks stock moves only weakly (ρ 0.059), and its mapping to scenario severity is a policy choice
- Event accuracy on news is preliminary: 0.94 F1 on tweets, 0.54 vs 0.49 on 120 LLM-labelled news items
- The bank is a scenario simulator: synthetic book, expert-set credit shocks, market moves calibrated
- Backtests trade next day on date-stamped text; live mode is a single-process prototype

Next steps

- Human-labelled news test set covering every event class
- Silver labels from the Groq-hosted teacher for credit and operational events; retrain the student
- Retrain the impact model on the improved event features
- Intraday timestamps for a latency-aware backtest; second-round contagion in Module B

Code, data sources and setup: public GitHub repository (README) · demo video linked there